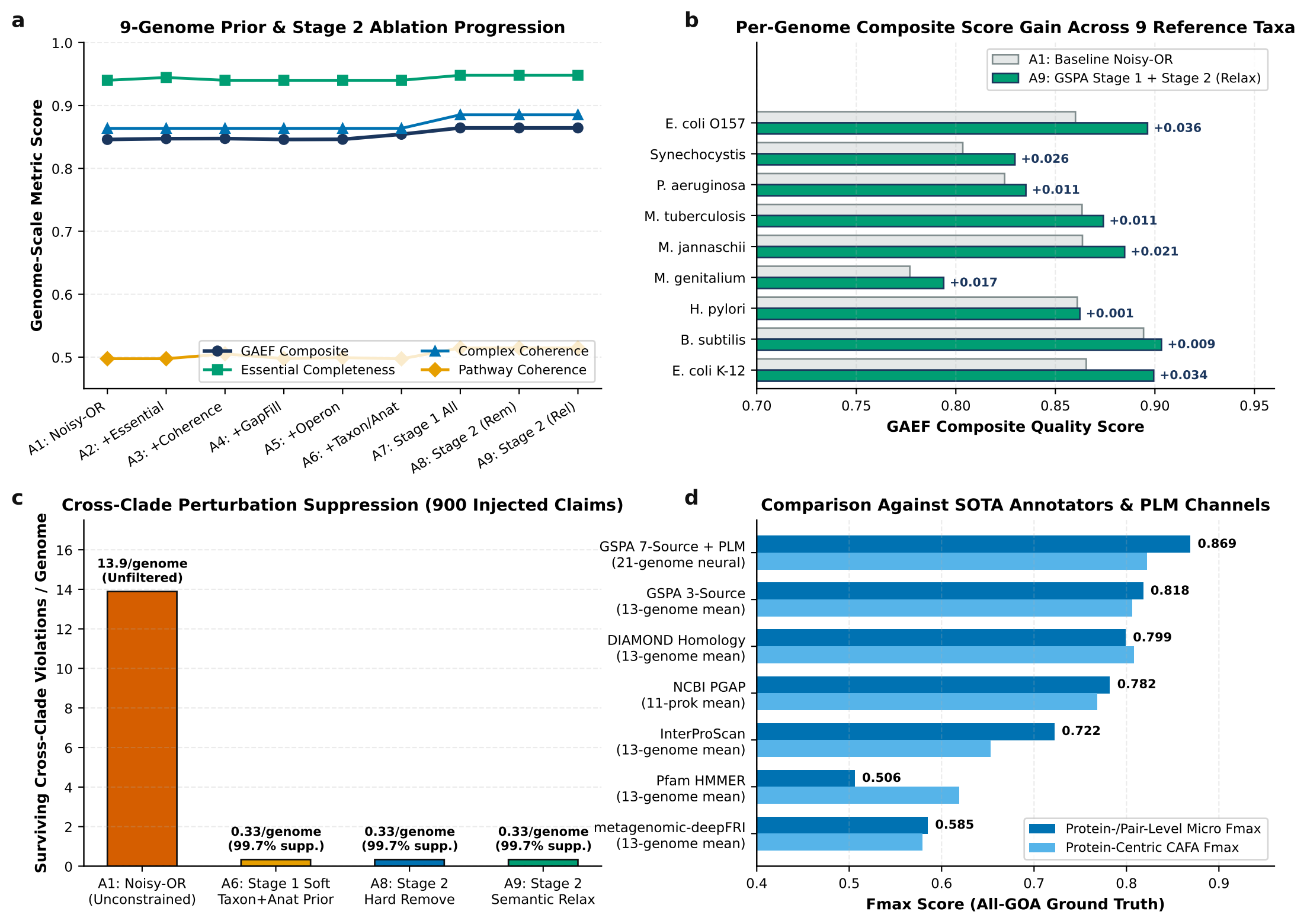
a
9-Genome Prior & Stage 2 Ablation Progression
Genome-Scale Metric Score
1.0
0.9
0.8
0.7
0.6
0.5
GAEF Composite
Essential Completeness
Complex Coherence
Pathway Coherence
A1: Noisy-OR
A2: +Essential
A3: +Coherence
A4: +GapFill
A5: +Operon
A6: +Taxon/Anat
A7: Stage 1 All
A8: Stage 2 (Rem)
A9: Stage 2 (Rel)
b
Per-Genome Composite Score Gain Across 9 Reference Taxa
A1: Baseline Noisy-OR
A9: GSPA Stage 1 + Stage 2 (Relax)
E. coli O157
Synechocystis
P. aeruginosa
M. tuberculosis
M. jannaschii
M. genitalium
H. pylori
B. subtilis
E. coli K-12
+0.036
+0.026
+0.011
+0.011
+0.021
+0.017
+0.001
+0.009
+0.034
0.70
0.75
0.80
0.85
0.90
0.95
GAEF Composite Quality Score
c
Cross-Clade Perturbation Suppression (900 Injected Claims)
Surviving Cross-Clade Violations / Genome
16
14
12
10
8
6
4
2
0
13.9/genome
(Unfiltered)
0.33/genome
(99.7% supp.)
0.33/genome
(99.7% supp.)
0.33/genome
(99.7% supp.)
A1: Noisy-OR
(Unconstrained)
A6: Stage 1 Soft
Taxon+Anat Prior
A8: Stage 2
Hard Remove
A9: Stage 2
Semantic Relax
d
Comparison Against SOTA Annotators & PLM Channels
GSPA 7-Source + PLM
(21-genome neural)
GSPA 3-Source
(13-genome mean)
DIAMOND Homology
(13-genome mean)
NCBI PGAP
(11-prok mean)
InterProScan
(13-genome mean)
Pfam HMMER
(13-genome mean)
metagenomic-deepFRI
(13-genome mean)
0.869
0.818
0.799
0.782
0.722
0.506
0.585
Protein-/Pair-Level Micro Fmax
Protein-Centric CAFA Fmax
0.4
0.5
0.6
0.7
0.8
0.9
Fmax Score (All-GOA Ground Truth)